

TRAINITY INSTAGRAM ANALYTICS PROJECT

Project Overview — I am a Data Analyst and I have to analyze user data and have to perform tasks such as **Loyal User Reward, Inactive User Engagement, Contest Winner Declaration, Hashtag Research, Ad Campaign Launch, User Engagement, Bots & Fake Accounts** using **MYSQL workbench**

- 1. TASK 1** - Loyal User Reward: The marketing team wants to reward the most loyal users, i.e., those who have been using the platform for the longest time.
Your Task: Identify the five oldest users on Instagram from the provided database.

APPROACH - We have to identify which are the top five users that have been using the Instagram accounts for the longest. For this, we will order the user by the time at which they created the account and then we will rearrange that in descending order and select the top 5 rows from it.

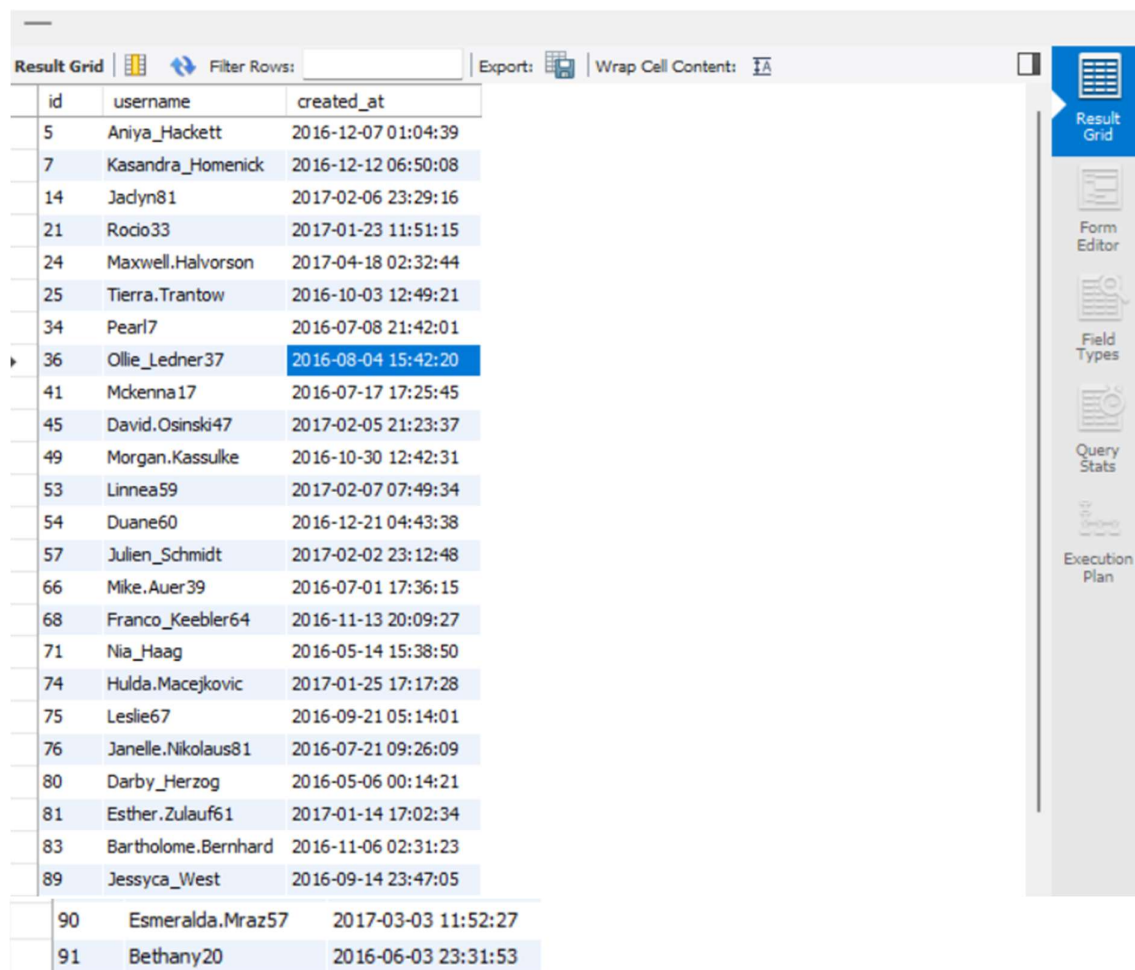
```
1 • USE Instagram_Database;
2 • Select id,username,created_at
3   FROM users
4   ORDER BY(created_at) DESC
5   LIMIT 5;
6
7
```

Result Grid			
	id	username	created_at
▶	11	Justina.Gaylord27	2017-05-04 16:32:16
	6	Travon.Waters	2017-04-30 13:26:14
	85	Milford_Gleichner42	2017-04-30 07:50:51
	19	Hailee26	2017-04-29 18:53:40
	24	Maxwell.Halvorson	2017-04-18 02:32:44
✱	NULL	NULL	NULL

2. TASK 2 - Inactive User Engagement: The team wants to encourage inactive users to start posting by sending them promotional emails. Your Task: Identify users who have never posted a single photo on Instagram

APPROACH – We have to identify the users who have never posted images on Instagram. For that we will count the users who have posted the photos on Instagram and by joining the tables users and photo we will remove the users that have posted on the Instagram from that join table

```
1 • use Instagram_Database;
2 • SELECT users_table.id, users_table.username, users_table.created_at FROM users users_table
3   LEFT JOIN photos photos_posted ON users_table.id = photos_posted.user_id
4   WHERE photos_posted.id IS NULL;
```



	id	username	created_at
	5	Aniya_Hackett	2016-12-07 01:04:39
	7	Kasandra_Homenick	2016-12-12 06:50:08
	14	Jadyn81	2017-02-06 23:29:16
	21	Rocio33	2017-01-23 11:51:15
	24	Maxwell.Halvorson	2017-04-18 02:32:44
	25	Tierra.Trantow	2016-10-03 12:49:21
	34	Pearl7	2016-07-08 21:42:01
▶	36	Ollie_Ledner37	2016-08-04 15:42:20
	41	Mckenna17	2016-07-17 17:25:45
	45	David.Osinski47	2017-02-05 21:23:37
	49	Morgan.Kassulke	2016-10-30 12:42:31
	53	Linnea59	2017-02-07 07:49:34
	54	Duane60	2016-12-21 04:43:38
	57	Julien_Schmidt	2017-02-02 23:12:48
	66	Mike.Auer39	2016-07-01 17:36:15
	68	Franco_Keebler64	2016-11-13 20:09:27
	71	Nia_Haag	2016-05-14 15:38:50
	74	Hulda.Macejkovic	2017-01-25 17:17:28
	75	Leslie67	2016-09-21 05:14:01
	76	Janelle.Nikolaus81	2016-07-21 09:26:09
	80	Darby_Herzog	2016-05-06 00:14:21
	81	Esther.Zulauf61	2017-01-14 17:02:34
	83	Bartholome.Bernhard	2016-11-06 02:31:23
	89	Jessyca_West	2016-09-14 23:47:05
	90	Esmeralda.Mraz57	2017-03-03 11:52:27
	91	Bethany20	2016-06-03 23:31:53

3. TASK 3 - Contest Winner Declaration: The team has organized a contest where the user with the most likes on a single photo win.

Your Task: Determine the winner of the contest and provide their details to the team.

APPROACH – We have to count the number of likes on photo and select the most liked photo on Instagram to giveaway prize of the user with the most liked single photo. For this we will take the users and photos and after that will count the number of likes on the photo. We will rearrange in descending order and select the 1st entry of it which will give photo and the contest winner with maximum likes on photo.

```
1 • use Instagram_Database;
2 • SELECT user_id, id AS photo_id, image_url,
3       (SELECT COUNT(*) FROM likes WHERE likes.photo_id = photos.id) AS like_count
4 FROM photos
5 ORDER BY like_count DESC
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	user_id	photo_id	image_url	like_count
▶	52	145	https://jarret.name	48

Result Grid
Form Editor

4. TASK - Hashtag Research: A partner brand wants to know the most popular hashtags to use in their posts to reach the most people.
Your Task: Identify and suggest the top five most commonly used hashtags on the platform.

APPROACH - We have to identify which are the top five hashtags most used on the platform. To accomplish this, we examine the photo_tags table that connects photos to hashtags, and count how many times each hashtag appears. Next, we join these counts with the actual hashtag names using the tags table. Finally, we rank them from most to least used and select the top five

```
1 • USE Instagram_Database;
2 • SELECT t.tag_name, COUNT(pt.photo_id) AS tag_count
3     FROM photo_tags pt
4     JOIN tags t ON pt.tag_id = t.id
5     GROUP BY t.tag_name
```

Result Grid		Filter Rows:	Export:	Wrap Cell Coi
tag_name	tag_count			
smile	59			
beach	42			
party	39			
fun	38			
concert	24			

5. Ad Campaign Launch: The team wants to know the best day of the week to launch ads.

Your Task: Determine the day of the week when most users register on Instagram. Provide insights on when to schedule an ad campaign.

APPROACH - First, we need to determine the best day to run an ad campaign for Instagram and analyze the day the majority of users created their accounts. Then, we will sum up their own results. It will help us decide when new users are active and engaged. We take a look at registration times for all users, and how many people signed up on each day of the week, to find the most popular day. That's the best day to show ads, because it's the time that user interest

```
1 • SELECT
2     DAYNAME(created_at) AS registration_day, COUNT(id) AS user_count
3 FROM users
4 GROUP BY registration_day
5 ORDER BY user_count DESC
6 LIMIT 1;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	registration_day	user_count			
▶	Thursday	16			

INVESTOR METRICS –

1. TASK – User Engagement: Investors want to know if users are still active and posting on Instagram or if they are making fewer posts.

Your Task: Calculate the average number of posts per user on Instagram. Also, provide the total number of photos on Instagram divided by the total number of users.

APPROACH - Investors want to know if users are still posting on Instagram or if engagement is slipping. So, to quantify this, we consider the average number of posts per user. Now we calculate the average number of photos uploaded by dividing the total number of photos uploaded on Instagram by the total number of users. A high number means users are actively posting. A low score could suggest a decline in engagement.

```
1 • SELECT
2     (COUNT(*) / (SELECT COUNT(*) FROM users)) AS avg_posts_per_user
3 FROM photos;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	avg_posts_per_user			
▶	2.5700			

2. TASK - Bots & Fake Accounts: Investors want to know if the platform is crowded with fake and dummy accounts.

Your Task: Identify users (potential bots) who have liked every single photo on the site, as this is not typically possible for a normal user.

APPROACH - Investors worry that Instagram could be populated with bots or fake accounts. In fact, one of the most obvious giveaways of these bots is if a user likes every single photo on the platform — something a normal person just wouldn't do. Then, to discover these suspicious user accounts, we check if the total likes given by the user is always less than the total number of photos, available on Instagram. If every single photo liked by a user might be a bot.

```
1 • SELECT user_id
2   FROM likes
3   GROUP BY user_id
4   HAVING COUNT(photo_id) = (SELECT COUNT(*) FROM photos);
5
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	user_id			
▶	5			
	14			
	21			
	24			
	36			
	41			
	54			
	57			
	66			
	71			
	75			
	76			
	91			

INSIGHTS – After working on this project I get to know about some of the criteria to analyze loyal users, one of the criteria to select a computerized id bots which was the user id who has liked all the photos. Also, I get to know how real-world insights is taken from databases.

RESULTS –

- TOP 5 LOYAL USERS OF INSTAGRAM ARE HAVING USER ID – 11, 6, 85, 19, 24
- THERE ARE 26 INACTIVE USERS.
- USER ID (52) HAVE MOST LIKED-ON PHOTO WITH 48 LIKES.
- TOP 5 HASHTAGS ON INSTAGRAM ARE SMILE, BEACH, PARTY, FUN, CONCERT.
- THURSDAY IS THE DAY WITH MOST USER REGISTRATION WHICH IS 16.
- AVERAGE INSTAGRAM USER POSTS 2.5700 POST PER USER.
- THERE EXIST 13 BOT ACCOUNTS POSSIBILITIES IN INSTAGRAM.

TECH USED – I HAVE USED MYSQL WORKBENCH 8.0 CE FOR THIS PROJECT.